

LIFTSYSTEM

AN ARDUINO-BASED ACCESSIBLE ELEVATOR CONTROL SYSTEM FOR PEOPLE WITH DISABILITIES

1.0 INTRODUCTION

LiftSystem is an Arduino-based accessible elevator control system designed to assist people with disabilities in moving between different floors safely and independently.

The system uses an Arduino Uno as the main controller and integrates several electronic components, including push buttons, a 16×2 LCD, a buzzer, an LED, a servo motor, an HC-SR04 ultrasonic sensor, a DC motor, and a transistor-based motor switching circuit.

The push buttons allow the user to select a destination floor. The LCD provides visual information about the elevator's current status and destination. The buzzer provides audio feedback when the elevator passes or reaches different floors. The servo motor is used to control the elevator door, while the ultrasonic sensor detects people or objects near the doorway.

An emergency button is also included to stop normal operation and activate an emergency alarm. A transistor is used as an electronic switching device between the Arduino and the DC motor because an Arduino digital output pin should not be used to drive a motor directly.

The project demonstrates the application of embedded systems, automation, electronic switching, sensor technology, and accessibility in the development of a small-scale elevator prototype.

2.0 PROBLEM STATEMENT

People with disabilities may experience difficulties when using conventional elevators, particularly when controls are difficult to operate or when information about the elevator's movement is provided through only one type of feedback.

For example, a visually impaired user may have difficulty determining when the elevator has reached a particular floor if information is provided only on a visual display. There is also a safety concern when an elevator door closes while a person or object is still in the doorway.

Therefore, there is a need for an accessible elevator control system that provides simple controls, multiple forms of feedback, and additional safety features.

LiftSystem was developed as a prototype to address these challenges by combining simple floor-selection buttons, visual LCD information, audio feedback, automatic door control, obstacle detection, and an emergency control mechanism.

3.0 OBJECTIVES

3.1 Main Objective

To design and implement an Arduino-based accessible elevator control system that provides simple operation, visual and audio feedback, automatic door control, and basic safety features for people with disabilities.

3.2 Specific Objectives

The specific objectives of the LiftSystem are:

1. To design an accessible elevator control system using an Arduino Uno.
2. To provide simple floor selection using push buttons.
3. To display elevator status and floor information using a 16×2 LCD.
4. To provide audio feedback using different buzzer tones for different floors.
5. To automatically control the elevator door using a servo motor.
6. To detect people or objects near the elevator doorway using an HC-SR04 ultrasonic sensor.
7. To provide an emergency button for stopping normal elevator operation.
8. To use a transistor as an electronic switching device for controlling the DC motor.
9. To design and test a complete prototype circuit in Tinkercad.
10. To demonstrate how embedded systems can be applied to improve accessibility and safety.

4.0 TOOLS AND COMPONENTS USED

The following tools and electronic components were used to design and implement the LiftSystem prototype.

4.1 Arduino Uno

The Arduino Uno is the main microcontroller of the LiftSystem. It processes input from the floor-selection buttons, emergency button, and ultrasonic sensor. It also controls the LCD, buzzer, LED, servo motor, and motor switching circuit.

4.2 Five Push Buttons

Five push buttons are used for selecting the destination floor:

- Button 1 – Basement
- Button 2 – Ground
- Button 3 – Floor 1
- Button 4 – Floor 2
- Button 5 – Floor 3

The buttons are configured using the Arduino's internal INPUT_PULLUP resistors. This reduces the need for external pull-up resistors.

4.3 16×2 LCD

A 16×2 LCD is used to provide visual information to the user.

The LCD can display messages such as:

- SYSTEM READY
- Added: Floor3
- Moving to: Floor1
- Moving to: Floor2
- Moving to: Floor3
- Arrived: Floor3
- Person detected
- !!! EMERGENCY !!!

The LCD allows the user to monitor the current state of the elevator.

4.4 Servo Motor

The servo motor controls the opening and closing of the elevator door.

When the elevator reaches the selected floor, the servo rotates to the programmed open position. After the doorway is clear, it rotates back to the closed position.

4.5 HC-SR04 Ultrasonic Sensor

The HC-SR04 ultrasonic sensor is used for obstacle detection near the elevator doorway.

The sensor measures the distance between itself and an object by transmitting an ultrasonic pulse and measuring the time taken for the echo to return.

When an object is detected within approximately **20 cm**, the system considers the doorway to be occupied. The LCD displays:

Person detected

The door remains open until the detected object moves away from the doorway.

4.6 Buzzer

The buzzer provides audio feedback to the user.

Different frequencies are assigned to different floors. For example:

Floor	Frequency
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Basement	600 Hz
----------	--------

Ground	700 Hz
--------	--------

Floor 1	800 Hz
---------	--------

Floor 2	900 Hz
---------	--------

Floor 3	1000 Hz
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The different tones allow the user to distinguish floor information through sound.

4.7 LED

An LED is used as a visual indicator during emergency operation.

When emergency mode is activated, the LED flashes together with the emergency buzzer to provide a visual warning.

4.8 Emergency Push Button

The emergency push button allows the user to activate emergency mode.

When the button is pressed, the system stops normal elevator movement, clears pending floor requests, activates the emergency alarm, and displays an emergency message on the LCD.

4.9 DC Motor

The DC motor represents the elevator movement mechanism in the prototype.

The motor is switched on when the elevator is simulated as moving between floors.

In a real elevator installation, the motor would require an appropriate motor controller, power supply, braking system, position feedback, and other safety mechanisms.

4.10 Transistor

A transistor is used as an **electronic switching device** to control the DC motor.

The Arduino digital output pin provides a control signal to the transistor rather than supplying the motor current directly.

For a small prototype motor, an NPN transistor such as a **2N2222** may be used. A resistor is connected between the Arduino output and the transistor base to limit the base current.

When the Arduino output is HIGH, the transistor switches ON and allows current to flow through the motor. When the Arduino output is LOW, the transistor switches OFF and stops the motor.

A flyback diode should be connected across the motor to protect the transistor from the voltage spike produced when the motor is switched OFF.

4.11 Resistors

Resistors are used for current limiting and protection.

In particular, a resistor can be used:

- In series with the LED to limit LED current.
- Between the Arduino output pin and the transistor base to limit base current.

4.12 Breadboard

The breadboard is used to construct the prototype circuit without soldering.

It provides a convenient method of connecting the Arduino, sensors, buttons, display, motor-control circuit, and other components.

4.13 Jumper Wires

Jumper wires are used to make electrical connections between the Arduino, breadboard, and individual components.

4.14 Power Supply

The Arduino provides power for suitable low-current components.

The DC motor should use an appropriate external power supply when its current requirement is higher than what the Arduino can safely provide.

The external motor supply and Arduino must share a common ground when the Arduino is controlling the transistor.

4.15 Tinkercad

Tinkercad is used to design and simulate the electronic circuit.

It allows the components to be connected virtually and provides a safe environment for testing the circuit before physical implementation.

5.0 COMPONENT CONNECTIONS

The main component connections are shown below.

Component	Arduino Connection
Basement button	Digital Pin 2
Ground button	Digital Pin 3
Floor 1 button	Digital Pin 4
Floor 2 button	Digital Pin 5
Floor 3 button	Digital Pin 6
Emergency button	Digital Pin 7
HC-SR04 Trigger	Digital Pin 8
HC-SR04 Echo	Digital Pin 9
Servo signal	Digital Pin 10

Component	Arduino Connection
Buzzer	Digital Pin 11
LED	Digital Pin 12
Motor control/transistor	Digital Pin 13
LCD RS	A0
LCD E	A1
LCD D4	A2
LCD D5	A3
LCD D6	A4
LCD D7	A5

The five floor-selection buttons are connected to digital pins 2–6, while the emergency button is connected to digital pin 7.

The HC-SR04 ultrasonic sensor uses pins 8 and 9 for the trigger and echo signals. The servo signal is connected to pin 10, the buzzer to pin 11, the emergency LED to pin 12, and the transistor control signal to pin 13.

The LCD uses analog pins A0–A5 as digital I/O pins for communication.

6.0 TRANSISTOR AND MOTOR CONTROL CIRCUIT

The transistor is an important part of the LiftSystem because it provides an interface between the Arduino and the DC motor.

An Arduino digital pin is designed to provide a low-current control signal. It should not be connected directly to a DC motor because the motor can require considerably more current and can generate voltage transients.

The transistor therefore operates as an electronic switch.

The simplified connection is:

Arduino Pin 13 → Base Resistor → Transistor Base

Transistor Collector → Motor Negative Terminal

Motor Positive Terminal → External Positive Supply

Transistor Emitter → GND

A flyback diode is connected across the motor:

Diode Cathode → Motor Positive

Diode Anode → Motor Negative

The Arduino ground and external motor supply ground must be connected together to provide a common electrical reference.

When Arduino pin 13 is set HIGH, the transistor turns ON and allows current to flow through the motor. When pin 13 is set LOW, the transistor turns OFF and interrupts the motor current.

Therefore, the transistor acts as a switch controlled by the Arduino.

7.0 SYSTEM DESIGN AND SCHEMATIC

The LiftSystem circuit schematic was designed using Tinkercad.

The schematic contains the following major sections:

1. Arduino Uno controller.
2. Five floor-selection buttons.
3. Emergency push button.
4. 16×2 LCD.
5. HC-SR04 ultrasonic sensor.
6. Servo motor for door control.
7. Buzzer for audio feedback.
8. LED for emergency indication.
9. DC motor representing elevator movement.
10. Transistor motor-switching circuit.
11. Resistors.
12. Breadboard and power connections.

The Arduino acts as the central controller. It receives input signals from the buttons and ultrasonic sensor, processes the information, and controls the output devices.

The transistor provides the electrical interface between the Arduino and the DC motor. This arrangement prevents the Arduino output pin from being used to supply the motor's operating current directly.

The schematic is useful during construction because it provides a clear representation of how the components are electrically connected.

8.0 WORKING PRINCIPLE

The LiftSystem operates through several stages.

8.1 System Initialization

When the system is powered on, the Arduino initializes the connected components.

The LCD displays the initial system status, and the elevator is set to start at the Ground floor.

The current floor is represented in the program using a variable such as:

```
currentFloor = 1
```

where the value represents the Ground floor in the programmed floor-index system.

8.2 Floor Selection

The user selects a destination by pressing one of the five floor buttons.

For example, when the user presses the Floor 3 button, the selected floor is added to the request queue.

The LCD displays:

Added:

Floor3

This confirms that the floor request has been received.

8.3 Elevator Movement

After a floor request is received, the system determines the direction and number of floors between the current position and the requested destination.

For example, if the elevator is at Ground and Floor 3 is selected, the simulated movement is:

Ground → Floor 1 → Floor 2 → Floor 3

The motor control output is activated during the simulated movement. The transistor receives the control signal from the Arduino and switches the motor circuit.

8.4 Floor Audio Feedback

As the elevator passes each floor, the buzzer produces the corresponding tone.

For example:

- Floor 1 → 800 Hz
- Floor 2 → 900 Hz
- Floor 3 → 1000 Hz

This provides audio feedback that can assist users who have difficulty reading the LCD.

8.5 Arrival at the Destination

When the elevator reaches the selected floor, the LCD displays the arrival message.

For example:

Arrived:

Floor3

The buzzer produces an arrival sound, and the servo motor is commanded to open the elevator door.

8.6 Door Safety Operation

After the door opens, the HC-SR04 ultrasonic sensor checks whether a person or object is present near the doorway.

If an object is detected within approximately 20 cm, the LCD displays:

Person detected

The system keeps the door open while the obstruction remains within the detection range.

When the doorway becomes clear, the servo motor closes the door.

This feature demonstrates the use of a sensor to improve the safety of an automated system.

8.7 Emergency Operation

The emergency button provides a mechanism for interrupting normal operation.

When the emergency button is pressed, the system enters emergency mode.

During emergency mode, the system:

- Stops normal motor operation.
- Clears pending floor requests.
- Displays !!! EMERGENCY !!! on the LCD.
- Activates the buzzer.
- Flashes the LED.
- Prevents normal elevator operation until the emergency condition is handled.

This provides both visual and audio emergency feedback.

9.0 SOFTWARE IMPLEMENTATION

The LiftSystem is programmed using **Arduino C++** in the Arduino IDE.

The program uses libraries and programming techniques for controlling the different components.

The main software components include:

9.1 Servo Control

The Servo.h library is used to control the servo motor responsible for opening and closing the elevator door.

9.2 LCD Control

The LiquidCrystal.h library is used to control the 16×2 LCD.

9.3 Floor Request Queue

An array-based queue is used to store selected floor requests.

This allows the system to process destination requests in an organized manner.

9.4 Current Floor Tracking

A variable such as `currentFloor` keeps track of the elevator's current position.

This information is used to determine the next movement step.

9.5 Buzzer Control

The `tone()` function can be used to generate different frequencies for the different floors.

9.6 Ultrasonic Distance Measurement

The HC-SR04 sensor is controlled by generating a trigger pulse and measuring the duration of the echo signal.

The measured time is converted into distance.

9.7 Emergency Handling

The program continuously checks the emergency button.

When an emergency is detected, normal movement is interrupted and the emergency response is activated.

The main program structure consists of:

- System setup.
- Main control loop.
- Floor request handling.
- Floor queue management.
- Elevator movement.
- Floor audio feedback.
- Arrival processing.
- Door control.
- Ultrasonic sensing.
- Emergency handling.

- Buzzer and LED control.

This modular structure makes the program easier to understand, test, and modify.

10.0 RESULTS

The LiftSystem prototype was designed and tested to demonstrate the intended elevator control functions.

The following results were achieved:

1. The system starts with the elevator positioned at the Ground floor.
2. Users can select a destination using the floor-selection buttons.
3. The selected destination is displayed on the LCD.
4. The elevator simulation moves between floors sequentially.
5. The LCD displays the current floor and movement status.
6. Different buzzer frequencies provide audio feedback for different floors.
7. The servo motor controls the opening and closing of the elevator door.
8. The ultrasonic sensor detects people or objects near the doorway.
9. The door remains open when an obstruction is detected.
10. The emergency button interrupts normal operation.
11. The emergency buzzer and LED provide warning signals.
12. The transistor provides a switching interface for controlling the DC motor.

For example, when Floor 3 is selected while the elevator is at Ground, the simulated movement is:

Ground → Floor 1 → Floor 2 → Floor 3

The system provides visual information through the LCD and audio information through the buzzer during this process.

11.0 IMPORTANCE OF THE SYSTEM TO PEOPLE WITH DISABILITIES

Accessibility is one of the main considerations in the LiftSystem design.

The system provides multiple forms of interaction and feedback.

The **push buttons** provide a simple method for selecting a destination floor.

The **LCD** provides visual information about the elevator's status.

The **buzzer** provides audio feedback for floor movement and arrival. This can assist users who have difficulty seeing the LCD.

The **ultrasonic sensor** provides basic doorway obstacle detection, helping prevent the door from closing while an object or person is detected.

The **emergency button** provides a simple mechanism for activating emergency operation.

By combining visual, audio, and physical controls, the system demonstrates how embedded technology can be designed with accessibility considerations.

12.0 CONCLUSION

The **LiftSystem** successfully demonstrates the design and implementation of an Arduino-based accessible elevator control system.

The prototype integrates an Arduino Uno, floor-selection buttons, a 16×2 LCD, buzzer, LED, servo motor, HC-SR04 ultrasonic sensor, DC motor, transistor, resistors, and other supporting components.

The system demonstrates several important functions, including floor selection, floor tracking, elevator movement simulation, audio and visual feedback, automatic door control, obstacle detection, and emergency operation.

The inclusion of a transistor in the motor-control circuit provides an appropriate electronic switching interface between the Arduino and the DC motor. This prevents the Arduino digital output from being used to directly supply the motor current.

Although LiftSystem is a prototype and is not intended for direct installation in a real building, it demonstrates important principles of embedded systems, automation, electronic switching, sensor integration, and accessibility.

A real elevator would require certified safety systems, industrial motor controllers, mechanical brakes, position sensors, redundant safety circuits, appropriate power systems, and compliance with relevant elevator safety standards.

13.0 FUTURE IMPROVEMENTS

The LiftSystem can be further improved in several ways.

13.1 Motor Driver

A dedicated motor driver or H-bridge can be used instead of a simple transistor switching circuit. This would allow better control of motor direction and speed.

13.2 Accurate Floor Position Detection

Position sensors, limit switches, rotary encoders, or other appropriate sensors could be added to determine the exact elevator position.

13.3 Voice Announcements

A voice module could be added to announce floor numbers and system status.

For example:

"Floor 3. Doors opening."

This would provide additional accessibility for visually impaired users.

13.4 Braille Buttons

Braille labels could be added to the floor buttons to make the control panel more accessible to visually impaired users.

13.5 Emergency Communication

A communication system could be added to allow a user to contact an operator or emergency service during an emergency.

13.6 Mobile Application

The system could be connected to a mobile application for monitoring, maintenance notifications, or remote system diagnostics.

13.7 Fault Detection

Additional software and sensors could be implemented to detect conditions such as motor failure, sensor failure, door failure, or abnormal movement.

13.8 Custom PCB

The breadboard prototype could be converted into a custom printed circuit board (PCB) for a more compact and reliable implementation.

13.9 Improved Power System

Separate regulated power supplies could be used for the Arduino, sensors, servo, and motor to provide more stable operation.

13.10 Additional Safety Sensors

Multiple safety sensors could be added to improve obstacle detection and system reliability.

14.0 PROJECT SUMMARY

Project Name: LiftSystem

Project Title: An Arduino-Based Accessible Elevator Control System for People with Disabilities

Project Type: Embedded System

Main Controller: Arduino Uno

Programming Language: Arduino C++

Development Platform: Arduino IDE and Tinkercad

Main Functions:

- Floor selection
- Floor movement simulation
- LCD status display
- Audio floor indication
- Automatic door control
- Obstacle detection
- Emergency operation
- Motor switching using a transistor

Main Components:

- Arduino Uno
- Five push buttons
- Emergency push button

- 16×2 LCD
- HC-SR04 ultrasonic sensor
- Servo motor
- DC motor
- NPN transistor
- Buzzer
- LED
- Resistors
- Breadboard
- Jumper wires
- Power supply

Application: Accessible elevator control prototype for people with disabilities.